Changing the arithmetic

Rings, fields, and vectors

$$X^2 - X - 1 = 0 \qquad \text{modulo } p$$

The matrix we already know

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$\det A = -1, \qquad A^2 = A + I.$$

Modulo any integer $m \geq 2$, the step is reversible.

Today: what arithmetic are these entries using?

Where are the roots?

$$\det(XI - A) = X^2 - X - 1.$$

Over the real numbers, its roots are

$$\frac{1+\sqrt{5}}{2}, \qquad \frac{1-\sqrt{5}}{2}.$$

Modulo 11: $4^2 = 5$.

Modulo 7: squares are $0, 1, 2, 4$.

Residues are numbers with arithmetic

$$\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \ldots, [n-1]\}.$$

$$[a] + [b] = [a+b], \qquad [a][b] = [ab].$$

For example, in $\mathbb{Z}/6\mathbb{Z}$,

$$[4] + [5] = [3], \qquad [4][5] = [2].$$

We usually omit the brackets once the modulus is fixed.

The ring rules

A commutative ring with identity has addition and multiplication.

- Addition is associative and commutative; it has 0 and negatives.
- Multiplication is associative and commutative; it has $1 \neq 0$.
- Multiplication distributes over addition.

$$a(b+c) = ab + ac.$$

A new phenomenon: zero divisors

In $\mathbb{Z}/6\mathbb{Z}$,

$$2 \cdot 3 = 0, \qquad 2 \neq 0, \quad 3 \neq 0.$$

A zero divisor is a nonzero element a for which

$$ab = 0 \quad \text{for some nonzero } b.$$

Cancellation can fail:

$$2 \cdot 1 = 2 \cdot 4, \qquad 1 \neq 4.$$

Which elements can we divide by?

A unit is an element u with a multiplicative inverse:

$$uv = 1 \quad \text{for some } v.$$

In $\mathbb{Z}/n\mathbb{Z}$,

$$[a] \text{ is a unit} \iff \gcd(a, n) = 1.$$

In $\mathbb{Z}/6\mathbb{Z}$, the units are 1 and 5.

Your turn: modulo eight

Work in $\mathbb{Z}/8\mathbb{Z}$.

1. Which of $1, 2, \ldots, 7$ are units?
2. Find the inverse of 3.
3. Find two nonzero elements whose product is 0.

Take one minute; then compare answers.

Modulo eight: answers

Units: $1, 3, 5, 7$.

$$3^{-1} = 3, \qquad 3 \cdot 3 = 9 = 1 \pmod 8.$$

$$2 \cdot 4 = 0 \pmod 8.$$

Being nonzero does not guarantee an inverse.

A field allows division

A field is a commutative ring with identity in which

every nonzero element has a multiplicative inverse.

Examples: $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{Z}/7\mathbb{Z}$.

A field has no zero divisors:

$$ab = 0,\ a \neq 0 \quad \Longrightarrow \quad b = a^{-1}0 = 0.$$

Why prime moduli give fields

$$\mathbb{Z}/n\mathbb{Z} \text{ is a field } \iff n \text{ is prime.}$$

If $n = p$ is prime, then $1 \leq a < p$ implies $\gcd(a, p) = 1$.

So every nonzero class has an inverse.

If $n = rs$ with $1 < r, s < n$, then

$$[r][s] = [0] \qquad \text{with } [r], [s] \neq [0].$$

We write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Characteristic: count copies of one

Write $m \cdot 1$ for the sum of m copies of 1.

The characteristic is the least positive m such that

$$m \cdot 1 = 0.$$

If no such m exists, the characteristic is 0.

$$\mathrm{char}(\mathbb{Q}) = 0, \qquad \mathrm{char}(\mathbb{Z}/n\mathbb{Z}) = n.$$

A field's positive characteristic is prime

Suppose the characteristic of a field is $m = rs$, with $1 < r, s < m$.

$$(r \cdot 1)(s \cdot 1) = m \cdot 1 = 0.$$

Minimality of m gives

$$r \cdot 1 \neq 0, \qquad s \cdot 1 \neq 0.$$

This contradicts the absence of zero divisors.

A field has characteristic 0 or a prime.

A finite field cannot have characteristic zero

In a finite field K, the list

$$0,\ 1,\ 2 \cdot 1,\ 3 \cdot 1, \ldots$$

must contain a repetition.

If $r \cdot 1 = s \cdot 1$ with $r < s$, subtract:

$$(s - r) \cdot 1 = 0.$$

Therefore $\mathrm{char}(K) = p$ for some prime p.

Every finite field contains a prime field

If $\mathrm{char}(K) = p$, the elements

$$0,\ 1,\ 2 \cdot 1, \ldots, (p-1) \cdot 1$$

form a copy of $\mathbb{F}_p$ inside K.

Addition and multiplication agree with arithmetic modulo p.

K may have more elements than these p elements.

Vectors and scalars

A vector space over a field k has

- vectors $u, v, \ldots$ that we can add;
- scalars $a, b, \ldots \in k$ that multiply vectors.

$$v \longmapsto av.$$

Examples: $\mathbb{R}^2$ over $\mathbb{R}$ and $\mathbb{F}_3^2$ over $\mathbb{F}_3$.

The vector-space rules

Vector addition has the usual associative and commutative laws, a zero vector, and additive inverses.

For $a, b \in k$ and $u, v \in V$,

$$a(u+v) = au + av, \quad (a+b)v = av + bv,$$
$$a(bv) = (ab)v, \qquad 1v = v.$$

The field k supplies the scalar arithmetic.

A familiar calculation over a different field

In $\mathbb{F}_3^2$, let $u = (1, 2)$ and $v = (2, 2)$.

$$u + v = (0, 1), \qquad 2u = (2, 1).$$

$$2u + v = (1, 0).$$

All coordinates are reduced modulo 3.

Same vector rules; different scalar arithmetic.

Linear combinations and span

A linear combination of $v_1, \ldots, v_r$ is

$$a_1 v_1 + \cdots + a_r v_r, \qquad a_i \in k.$$

Their span is the set of all such combinations.

In $\mathbb{F}_3^2$, for $u = (1, 2)$,

$$\operatorname{span}(u) = \{(0, 0), (1, 2), (2, 1)\}.$$

Your turn: a span over $\mathbb{F}_2$

In $\mathbb{F}_2^2$, let $u = (1, 1)$ and $v = (1, 0)$.

1. List the vectors in $\mathrm{span}(u)$.
2. List all vectors $au + bv$ with $a, b \in \mathbb{F}_2$.
3. Can $(0, 1)$ be written as $au + bv$?

Remember: the only scalars are 0 and 1.

Four coefficient choices, four vectors

(a, b)	$au + bv$
$(0, 0)$	$(0, 0)$
$(1, 0)$	$(1, 1)$
$(0, 1)$	$(1, 0)$
$(1, 1)$	$(0, 1)$

$$\operatorname{span}(u) = \{(0, 0), (1, 1)\}, \qquad (0, 1) = u + v.$$

What we have built today

- Residue arithmetic gives rings; prime moduli give fields.
- Every finite field contains some $\mathbb{F}_p$.
- Vector spaces can use $\mathbb{F}_p$ as their scalar field.

Next time: bases, counting, and a field with 49 elements.

Exit question

Why is $\mathbb{Z}/6\mathbb{Z}$ a ring but not a field?

Why can it not serve as the scalar *field* in our definition of a vector space?

Give a specific arithmetic obstruction.

Rings as scalars: a glimpse beyond vector spaces

One can use a ring of scalars and keep the vector-space identities.

The resulting object is called a module.

Over $\mathbb{Z}/6\mathbb{Z}$, $2v = 0$ need not imply $v = 0$:

$$v = 3, \qquad 2v = 0.$$

Our next lectures use fields, where nonzero scalars are invertible.

One set, two different scalar fields

The complex numbers form a vector space over $\mathbb{R}$:

$$z = a + bi, \qquad a, b \in \mathbb{R}.$$

The same set forms a vector space over $\mathbb{C}$:

$$z = z \cdot 1.$$

Changing the scalars can change the dimension.